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Revolutionizing Agriculture: Smart Plant Disease Detection with Deep Learning for Sustainable Farming

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Abstract—Adopting state-of-the-art techniques is necessary due to the necessity of early and accurate plant disease identification in agriculture. The effectiveness of ResNet-50, a cutting-edge convolutional neural network architecture, in image analysis-based plant disease diagnosis is examined in this work, this work makes use of a large dataset that covers a broad range of different types of plant illnesses. The ResNet-50 model, which was pre-trained on ImageNet and carefully adjusted to the domain-specific dataset, performs exceptionally well in disease classification tasks. This paper investigates optimization strategies, the difficult training process, and the trade-off between predictive accuracy and model complexity. Findings demonstrate the model's exceptional accuracy and resilience, outperforming traditional diagnostic techniques in terms of speed and dependability. These results highlight the revolutionary potential of ResNet-50 in transforming farming methods, facilitating early intervention, and lowering crop losses. 96,206 images of both healthy and diseased plant leaves make up the dataset, which represents the 38 most common disease classes in 14 different crop species.

Keywords—Plant Disease Detection, ResNet-50, Deep Learning, Image-Based Classification, Precision Agriculture, Crop Health Monitoring, Sustainable Farming, Early Disease Diagnosis, Agricultural Technology, Smart Farming Solutions, Plant Pathology, Crop Yield Optimization

I. INTRODUCTION

The pressures of a growing global population, climate change, and food security concerns are driving a revolutionary shift in agriculture, the foundation of human survival. Plant diseases are one of the many issues affecting the industry, they can cause large crop losses, lower yields, and unstable economies, especially in areas where agriculture is a major industry [7]. plant diseases seriously hinder agricultural output, resulting in financial losses and food supply risks. To reducing the consequences of plant leaf diseases early and accurate identification is essential, enabling timely interventions, and enhancing agricultural operations. This work promotes to enhance the crop health and productivity using AI-driven agricultural solutions by offering a high scalability and efficiency. By laying the foundation for the integration of deep learning into real-world agricultural applications, this work promotes global food security and opens the door to more sustainable farming methods. Traditional plant disease management techniques, which mostly rely on expert analysis and physical observation, are often labour-intensive, inefficient, and inadequate to meet the demands of modern agriculture. Additionally, by allowing diseases to develop unchecked, these

strategies run the danger of postponing intervention, which would increase their burden [3]. A revolutionary shift in agriculture, by the pressures of a growing global population the foundation of human survival is being driven, climate change, and food security concerns. One area of artificial intelligence, deep learning, has garnered a lot of attention because it has the potential to revolutionize the identification of plant diseases. Deep learning, particularly with Convolutional Neural Networks (CNNs), features that the human eye might miss can analyze by large volumes of visual data and identify complex patterns. This feature is crucial in the context of plant disease identification, where symptoms may be subtle, overlap, or physically identical across illnesses and crop kinds. This work focuses on implementing ResNet-50, a state-of-the-art CNN architecture that is renowned for its exceptional accuracy in tackling difficult image classification tasks. The decrease in performance in deeper networks is one of the primary problems in deep learning and it addressed by ResNet-50 by implementing residual learning techniques. Its ability to extract important features from high-resolution plant pictures makes it an extremely effective model for accurately identifying diseases, even in challenging circumstances [11]. In this work, we use ResNet-50 to provide a scalable and robust framework for detecting and classifying plant diseases in various crops. Using such sophisticated disease detection technologies has a lot to offer agriculture.

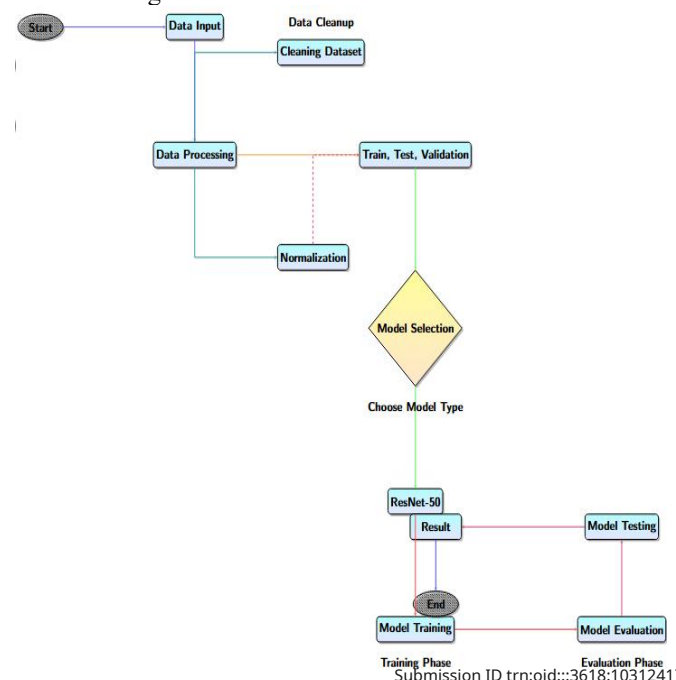


Figure 1: Deep Learning Model Flow Chart

Table

1:

SI No.	Researchers	Detection Algorithms	Images (Dataset)	Evaluating Parameters
1.	Shubham Malvankar et al. (2020) [10]	Naïve Bayes Algorithm, K-Nearest Neighbor Algorithm	500 images	Naïve Bayes Classifier has an accuracy of 65%, K Nearest Neighbor has an average accuracy of 80 %.
2.	Meenalochini M et al. (2024) [11]	GNN classification	750 Images	Average accuracy of 89%
4.	Md. Manowarul Islam et al. (2023) [14]	CNN, VGG-16, VGG- 19, and ResNet-50 architectures	1,000 Images	Accuracy rate CNN - 98.60%, VGG-16 - 92.39%, VGG- 19 - 96.15%, ResNet-50 - 98.98%.
6.	Giovanni Delnevo et al. (2022) [16]	Convolutional Neural networks (CNN)	2598 images	Accuracy rate DenseNet121 - 94.51%, MobileNet - 93.17%, MobileNetV2 - 94.58%, NASNetMobile - 93.97%.
7.	Rahul Sharma et al. (2021) [17]	CNN model	5932 images	Accuracy of 97.00%
8.	R. Sujatha et al. (2021) [20]	Support Vector Machine (SVM), Random Forest (RF), Stochastic Gradient Descent (SGD)) & DL (Inception-v3, VGG-16, VGG-19)	609 images	RF-76.8%, SGD-86.5%, SVM- 87%, VGG-19-87.4%, Inception-v3-89%, VGG-16-89.5%.

Comparison of Plant Disease Detection Algorithms, Datasets, and Evaluation Metrics

SI. No	Culture	Journal Name and Publication Year	Title	Technology	Identify Diseases
1.	Ginger [2]	International Conference on Data Science and Network Security (ICDSNS) [2023]	Revolutionizing Crop Management: An Emphasis on Ginger Leaf Disease Detection Techniques using Machine Learning and IoT	CNN, MOBILENETV2, AND VGG-16	healthy and an unhealthy leaf
2.	Ginger [3]	Mesay Gameda Yigezu on 29	Ginger Disease Detection Using a	Computer Vision Pre-Trained	Healthy ginger leaf and bacterial

Table 2: Research on Plant Leaf Disease Detection: Technologies, Studies, and Identified Diseases

		September [2023]	Computer Vision Pre-Trained Model	Model	wilt ginger leaf	
43	3.	Ginger [4]	4th International Conference on Innovation Practices in Technology and Management (ICIPTM) [2024]	Hybrid CNN & Random Forest Model for Effective Ginger Leaf Disease Diagnosis	CNN & Random Forest Model	Leaf Spot, Yellow Leaf Roll, Leaves The blight, bacterial Wilt, Rhizome Rot, Fusarium Wilt, The powdery Mildew, and Anthracnose Bacterial Leaf Blight and Brown Spot
24	4.	Rice [5]	2nd International Conference on Innovative Practices in Technology and Management (ICIPTM), Gautam Buddha Nagar, India [2022]	Automatic Rice Leaf Disease Segmentation Using Image Processing Techniques	K-means clustering	
25						<i>A. Plant Disease Detection Using ResNet-50</i>
26	5.	Apple [6]	Disease Detection in Apple Leaves Using Image Processing Techniques", Eng. Technol. Appl. Sci. Res., vol. 12, no. 2, pp. 8335-8341, Apr. [2022]	Disease Detection in Apple Leaves Using Image Processing Techniques	CNN, SVM, and KNN	healthy or diseased

networks is made possible by ResNet-50's use of a deep residual learning architecture, which solves issues like disappearing gradients. This architecture is very good at correctly diagnosing plant diseases while preserving computing efficiency because it is especially skilled at extracting important information from high-resolution photos.

II. PLANT DISEASES USED FOR THE MODEL

Numerous plant sections are impacted by **plant diseases**, which are brought on by pathogens such as **bacteria**, viruses, **fungi**, and environmental stressors. Prompt discovery is necessary for efficient management.

The following lists common plant diseases along with their traits and ways to treat them:

- A. **White powdery fungus:** that causes curling and browning on leaves, stems, and fruits is called powdery mildew. Fungicides, appropriate spacing, and avoiding overhead irrigation are used to manage it.
- B. **Downy mildew:** a disease characterized by white or yellow patches underneath leaves and that thrives in damp conditions and it is controlled with fungicides, air movement, and resistant cultivars.
- C. **Rusts:** rust-colored patches on stems and leaves that are controlled with fungicides and resistant plants;
- D. **Blights:** dark patches on leaves, stems, and fruits that can be prevented by fungicides, crop rotation, and resistant cultivars;
- E. **Bacterial Spots:** These tiny, dark spots on leaves and fruits can be avoided by avoiding overhead watering, applying copper-based treatments, and raising seedlings free of pathogens; and
- F. **Cucumber Mosaic Virus:** Aphid treatment, plant removal, and planting virus-free stock are methods of controlling this virus, which results in mosaic-like leaf patterns and stunted growth.
- G. **Aster Yellows:** Controlled by lowering leafhopper numbers and eliminating afflicted plants, this disease results in yellowing, reduced growth, and aberrant sprouts.

A. Description

Table 1: Comparative Literature Survey Table The results of the literature review must be summarized using a comparison table, particularly when assessing how well different plant disease prediction technique function. An overview of Table 3's content, including its assessment parameters and dataset sizes (number of photos), is provided below.

Table 2: Evaluation of Deep Learning Techniques for Predicting Plant Diseases A thorough comparison of the several deep learning techniques utilized in plant disease prediction is given in Table 2. It provides a clear overview of the advantages and disadvantages of various techniques by highlighting the diseases that have been recognized, the methodologies that have been used, and the corresponding performance measures. To assess the efficacy of current approaches and pinpoint possible areas for future research.

B. Developments in Deep Learning

A survey of labeling and categorization using leaf concepts is bestowed in this place item [1]. Using SVM, Deep Convolution NN and Nearest Neighbor can control the amount of pesticides and different compounds used for ailment control. Identification of spirit leaf ailment [2]. By utilizing K-resources assembling, support vector motor (SVM), convolutional affecting animate nerve organs networks (CNN) and usual representation handle methods, ailments maybe discovered and brand administration can be enhanced. Identification of spirit ailment [3]. MobileNet V2 pre-prepared models and progressive data accumulation are used to defeat convert opportunity and increase user knowledge. Good disease of spirit leaf spot affliction [4]. Random thicket means and a convolutional neural network (CNN) are used to without thinking categorize ailments and judge conduct metrics. Automatic categorization of edible grain leaf ailment [5]. SVM is used to increase veracity by thinking the presence of afflictions and without thinking pinpoint ailments. Disease of sphere leaf blight [6]. GoogleNet uses CNN, SVM and KNN to detect and bar the spread of ailments. Automatic discovery and reasoning of leaf blight utilizing a predicting model [7]. In order to determine the types of plant ailments that are extending, categorization designs and part ancestry algorithms are used. Diagnosis of plant leaf affliction [8]. AlexNet and ResNet-50 architectures are used to upgrade the adeptness of the invulnerable whole. A review of image treats methods for plant leaf affliction indicator [9]. Improving the veracity of plant disease disease and lowering the use of pesticides is attained by utilizing K-resources, SVM, ANN and fuzzy categorization orders. An intelligent. IOT and machine learning for smart agriculture [10]. The support vector machine feeds the current soil conditions and predicts and identifies the type of crop that will grow best in the agricultural field by suggesting the best conditions for the current soil conditions. Forecast for Cauliflower Plant Diseases [11] Principal Component Analysis (PCA), Random Forest (RF), Decision Trees (DT), Convolutional Neural Networks (CNN), Deep Neural Networks (DNN), Linear Discriminant Analysis (LDA), Automated Disease Detection, and IoT and Deep Learning Integration improving feature extraction, integrating with other technologies, and growing the dataset. Drone-captured photographs of both healthy and sick crops [12] CNN stand for Convolutional Neural Network. Real-time monitoring, early disease detection, and the development of extensive data increasing the robustness of the model, integrating it with decision support systems, and investigating other

machine learning methods. LSTM Network-Based Plant Disease Prediction [13] printed circuit board, Wi-Fi module, and low-cost LWS Early Illness Identification To increase the accuracy of plant disease predictions, we will extend our study by installing more sensor nodes and concurrently taking plant photos for a full crop cycle. We will then combine the two approaches, using both field-collected sensor data and plant images. Web-based program for crop disease prediction [14] CNN, ResNet-50, VGG-16, and VGG-19 forecast of crop diseases enhance the accuracy rate by creating a novel hybrid deep-learning architecture that combines plant leaf disease area identification or a localization technique to pinpoint the diseased leaf's location with an attention-based mechanism. Identifying Leaf Diseases in Plants [15] Using deep learning methods, a mobile-based system that uses a CNN model with two convolutional layers can automatically diagnose plant leaf diseases. To better anticipate the less performing illness groups and investigate the resilience of our method on a larger scale, trials with larger datasets are required. Method for Predicting Plant Diseases [16] To evaluate user interface usability and engagement levels, the Convolutional Neural Network (CNN), System Usability Scale (SUS), and User Engagement Scale (UES) are utilized. To enable more effective management of sensor data, separate the collection of sensor data from data aggregation and analysis. Examine in greater detail the benefits of utilizing a SIoT architecture, taking use of the plants' close proximity, and enhancing the prediction model's effectiveness.

Identification of Plant Diseases and Classification of Images [17] A deep learning model for the diagnosis of plant diseases, with a particular focus on coffee leaves, is developed utilizing the Support Vector Machine (SVM), K-Nearest Neighbors (KNN), Decision Tree, and Random Forest methods. This entails training the model with a dataset that records different plant health conditions and stages. Hyper-parameters of the proposed CNN model, such as the number of epochs, batch size, number of filters in each convolution layer, size of the filters, number of convolution layers, activation function for the convolution layer, pooling after each convolution layer, number of fully connected layers, neurons per layer, etc., can be optimized using search and optimization algorithms inspired by nature. Xception and MobileNetV2 models for plant disease prediction [18] By combining the MobileNetV2 and Xception models in an ensemble method, the Lite Ensemble MobileNetV2 and Xception LEMOXINET model aims to improve the identification of plant diseases. The CNN Xception or Extreme Inception model provides an enhanced version of the classic Inception. Using human or automated augmentation approaches to add datasets will increase the performance of leaf disease identification. Using machine learning to detect plant leaf diseases [19] Random Forest (RF), Squeezenet (SN), Linear-support vector machine (L-SVM), Stochastic gradient descent (SGD), Inception-V3, and VGG-16. Optimizing Farming Methods. propose to use bio-inspired techniques and fuzzy logic, which could have a big impact if the system's CA improves even with a tiny dataset. Predicting Diseases Through the Use of Ensembling Machine Learning [20] CNN model: Classification of Disease Severity, Prediction of Disease, and Assistance for Farmers In addition to forecasting a plant leaf illness, they can also be grouped according to the severity of the disease based on its symptoms. The method would be more beneficial if farmers were given suggestions for various disease-curing techniques and their availability.

C. Deep learning in farming

Deep learning applications in agriculture are a relatively

new trend. Research has shown CNNs' capacity for image-based plant disease detection. Aspects including model selection, data augmentation, transfer learning, and fine-tuning these models on datasets that include pictures of both healthy and ill plants have all been the subject of studies. As demonstrated by the results, models such as ResNet-50 can classify plant diseases with high accuracy, providing notable advancements over conventional detection techniques.

IV. METHODOLOGY

A. Dataset overview

96,206 images of both healthy and diseased plant leaves make up the dataset, which represents the 38 most common disease classes in 14 different crop species. ([Agro-Disease-Detector/PlantVillage at master · ahmed-pvamu/Agro-Disease-Detector · GitHub](#))

Distribution of Leaf Conditions (Total Images: 118)

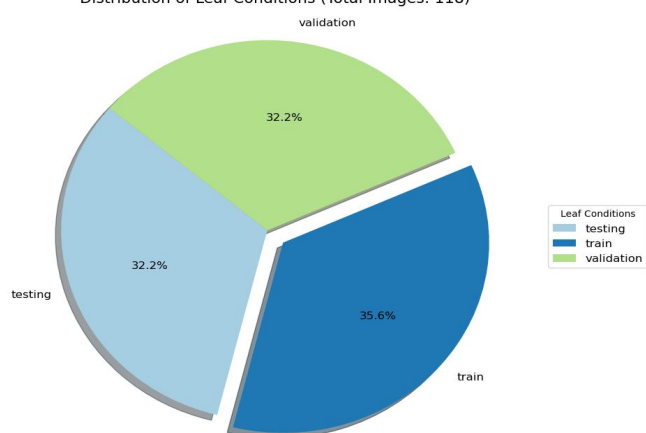


Figure 2: Distribution of Leaf Conditions in training, test and validation set.

Overall Leaf Condition Distribution (Healthy vs Diseased)
Healthy Leaves: 2681, Diseased Leaves: 5558

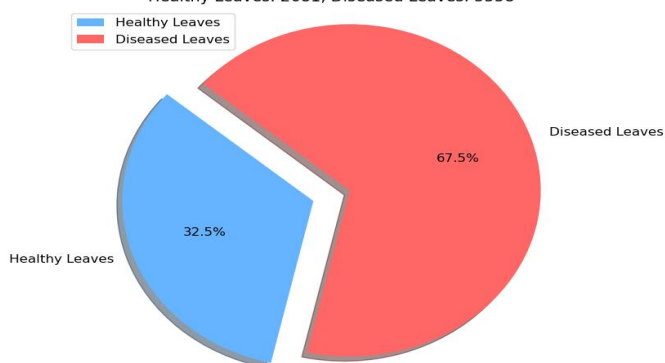


Figure 3: Graph which specifies Healthy and Diseased leaves.

B. Preprocessing Steps

All photos were resized to a uniform 224x224 pixel size as part of the preprocessing for the ResNet-50V2 model. Data augmentation methods like rotation, flipping, and zooming were used to reduce overfitting and improve the model's capacity for generalization, while pixel values were standardized to improve model performance.

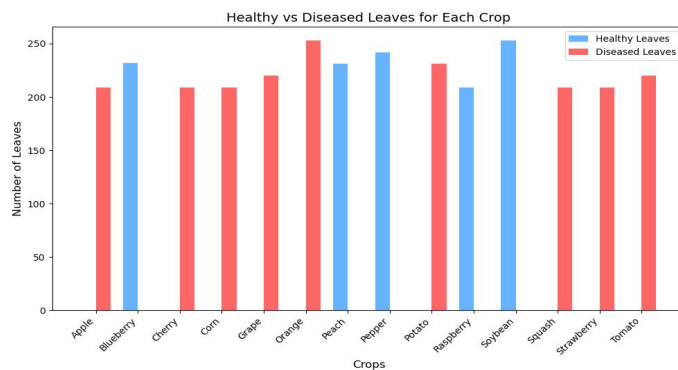


Figure 4: Healthy vs Diseased Leaves – Bar graph

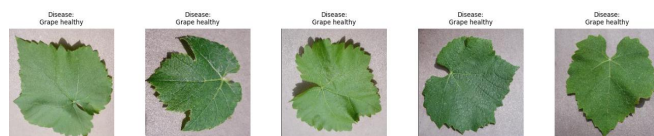


Figure 5: Feature selection dataset sample - Grape healthy

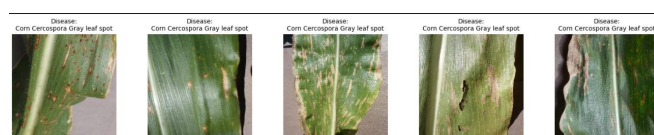


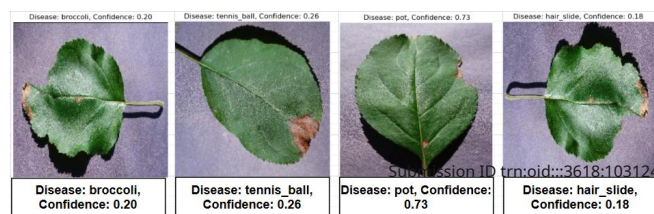
Figure 6: Feature selection dataset sample - Corn Cercospora Gray leaf spot



Figure 7: Feature selection dataset sample – Apple Black rot

C. Using ResNet50 to Classify Plant Leaf Images

By extracting rich information from input photos, the deep convolutional neural network ResNet50 is used to identify images of plant leaves. In order to find illness patterns, the procedure entails scaling photos to 224x224 pixels, standardizing pixel values, and using transfer learning. By ensuring precise classification, this method facilitates better agricultural practices and efficient plant disease diagnostics.



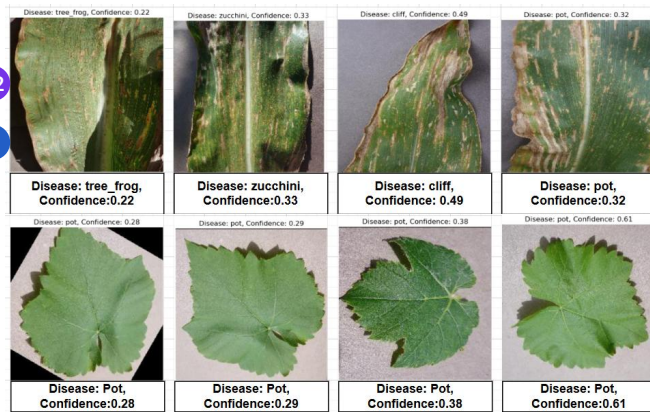


Figure 8: Image classification using ResNet50

Figure 8 Describe The probability or certainty that a model allocates to each predicted class for a specific input image is known as image classification confidence. The model's level of confidence in its forecast for a given class is represented by a numerical value, usually between 0 and 1 (or expressed as a percentage between 0% and 100%).

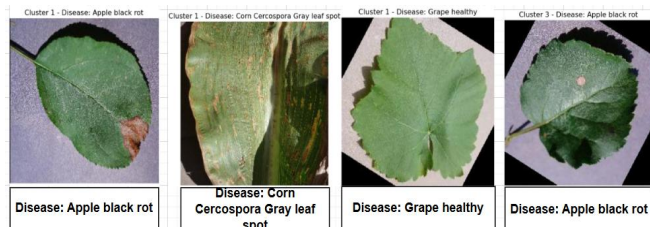


Figure 9: Feature Extraction

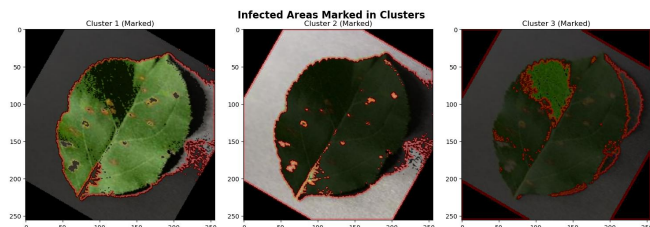


Figure 10: Visualization of Plant Leaf Disease Detection using Heatmaps

Figure 10 Describe The graphic highlights areas impacted by illness by comparing a raw plant leaf image with the associated heatmap visualizations. The model's emphasis on crucial leaf regions for precise diagnosis is validated by these visualizations, which also aid in spotting illness trends.

Figure 11 Describe A plant leaf (top-left) with obvious disease spots is shown in this image, followed by heatmap visualizations (ACB-2 to ACB-5) produced by deep learning models. These visualizations help interpret the model's attention patterns by effectively identifying disease-affected areas on the leaf surface, the highest attention indicating with red areas and less significance indicating by blue areas. The bottom row presents finer-grained heatmaps with more contrast and detail to validate the accuracy of illness classification models.

VI. CONCLUSION

This work makes a strong argument for the application of ResNet-50 in plant disease diagnosis, in addition to highlighting the technology's importance in promoting environmentally friendly agricultural methods, in order to strengthen farmers, improve crop health, and safeguard global

food chains, this endeavour attempts to open the door for creative solutions by fusing cutting-edge methods with the demands of contemporary agriculture. 96,206 images of both healthy and diseased plant leaves make up the dataset, which represents the 38 most common disease classes in 14 different crop species. The findings not only raise the possibility of tackling current agricultural issues, but they also encourage more in-depth conversations on the ways in which new technology could support a sustainable future. Furthermore, this work shows how cutting-edge technology may bridge the gap between agriculture and technology to address significant global issues like food security and sustainability. Large-scale disease detection gives farmers effective tools for decision-making, increasing output while protecting natural resources. From a wider angle, this work encourages more research into the integration of technology with mobile applications for greater accessibility and IoT devices for real-time disease monitoring. These developments have the potential to democratize access to technology, enabling small-scale farmers all across the world to use disease detection technologies.

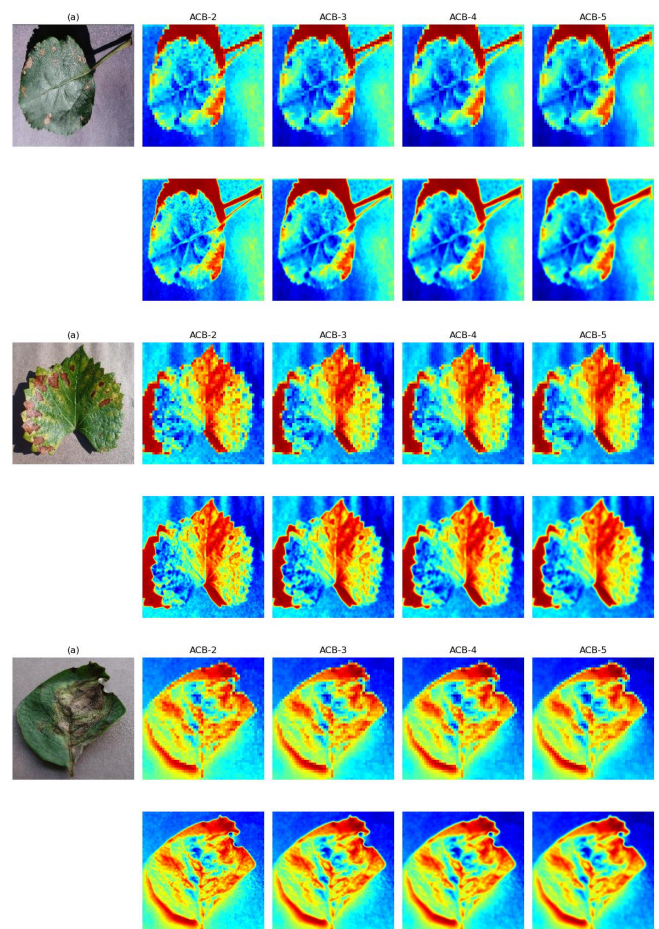


Figure 11: Heatmap Analysis for Disease-Spot Detection on Plant Leaf

V. RESULTS

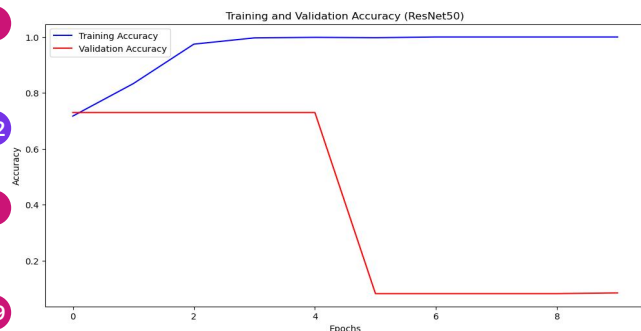


Figure 12: Training and validation Accuracy

Figure 12 Describe Average Training Accuracy = ~95%, Average Validation Accuracy = ~35%

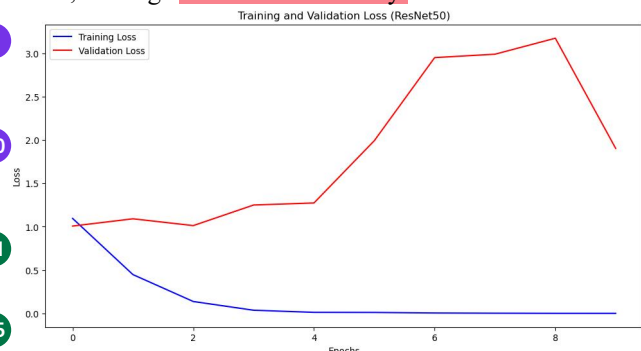


Figure 13: Training and validation loss

Figure 13 Describe Average Training Loss = ~0.2, Average Validation Loss = ~2.

REFERENCES

- [1] Kaur, S., Pandey, S. & Goel, S. Plants Disease Identification and Classification Through Leaf Images: A Survey. *Arch Computat Methods Eng* 26, 507–530 (2019).
- [2] S. C. M and C. Raju, "Revolutionizing Crop Management: An Emphasis on Ginger Leaf Disease Detection Techniques Using Machine Learning and IoT," *2023 International Conference on Data Science and Network Security (ICDSNS)*, Tiptur, India, 2023, pp. 1-5.
- [3] Kolesnikova, O., Yigezu, M.G., Tonja, A.L., Woldeyohannis, M.M., Sidorov, G., Gelbukh, A. (2023). Ginger Disease Detection Using a Computer Vision Pre-Trained Model. In: Rivera, G., Rosete, A., Dorronsoro, B., Rangel-Valdez, N. (eds) *Innovations in Machine and Deep Learning. Studies in Big Data*, vol 134. Springer, Cham
- [4] D. Banerjee, N. Sharma, D. Upadhyay, S. Dangi, N. Yamsani and K. S. Gill, "Hybrid CNN & Random Forest Model for Effective Ginger Leaf Disease Diagnosis," *2024 4th International Conference on Innovative Practices in Technology and Management (ICIPTM)*, Noida, India, 2024, pp. 1-6, doi: 10.1109/ICIPTM59628.2024.10563434.
- [5] Ks, Archana & Sahayadhas, Arun. (2018). Automatic Rice Leaf Disease Segmentation Using Image Processing Techniques. *International Journal of Engineering and Technology (UAE)*. 7. 182-185. 10.14419/ijet.v7i3.27.17756.
- [6] Alqethami, S., Almtanni, B., Alzhrani, W. and Alghamdi, M. 2022. Disease Detection in Apple Leaves Using Image Processing Techniques. *Engineering, Technology & Applied Science Research*. 12, 2 (Apr. 2022), 8335–8341.
- [7] N. Goel, D. Jain and A. Sinha, "Prediction Model for Automated Leaf Disease Detection & Analysis," *2018 IEEE 8th International Advance Computing Conference (IACC)*, Greater Noida, India, 2018, pp. 360-365.
- [8] H. Ajra, M. K. Nahar, L. Sarkar and M. S. Islam, "Disease Detection of Plant Leaf using Image Processing and CNN with Preventive Measures," *2020 Emerging Technology in Computing, Communication and Electronics (ETCCE)*, Bangladesh, 2020, pp. 1-6.
- [9] Cabanac G, Labbe C, Magazinov A, 2021, Tortured phrases: A dubious writing style emerging in science. Evidence of critical issues affecting established journals.
- [10] Malvankar, Shubham. (2020). "Plant Disease Detection using Image Processing." *International Journal for Research in Applied Science and Engineering Technology*. 8. 2685-2690. 10.22214/ijraset.2020.5450.

- [11] M., M., & Amudha, P. (2024). GNN Based Cauliflower Plant Disease Prediction Using Deep Learning Techniques. *International Journal of Intelligent Systems and Applications in Engineering*, 12(20s), 859–869. Retrieved from <https://ijisae.org/index.php/IJISAE/article/view/5313>
- [12] Vardhan, J. Vishnu and Kothapalli Sai Swetha. "Detection of healthy and diseased crops in drone captured images using Deep Learning." *ArXiv abs/2305.13490* (2023): n. pag.
- [13] K. S. Patle, R. Saini, A. Kumar and V. S. Palaparthi, "Field Evaluation of Smart Sensor System for Plant Disease Prediction Using LSTM Network," in *IEEE Sensors Journal*, vol. 22, no. 4, pp. 3715-3725, 15 Feb. 15, 2022, doi: 10.1109/JSEN.2021.3139988.
- [14] Islam, Md. Manowarul et al. "DeepCrop: Deep learning-based crop disease prediction with web application." *Journal of Agriculture and Food Research* (2023): n. pag.
- [15] Ahmed, Ahmed Abdelmoamen, and Gopireddy Harshavardhan Reddy. 2021. "A Mobile-Based System for Detecting Plant Leaf Diseases Using Deep Learning" *AgriEngineering* 3, no. 3: 478-493. <https://doi.org/10.3390/agriengineering3030032>
- [16] G. Delnevo, R. Girau, C. Ceccarini and C. Prandi, "A Deep Learning and Social IoT Approach for Plants Disease Prediction Toward a Sustainable Agriculture," in *IEEE Internet of Things Journal*, vol. 9, no. 10, pp. 7243-7250, 15 May 15, 2022, doi: 10.1109/JIOT.2021.3097379.
- [17] Sharma, Rahul, Amar Singh, N. Z. Jhanjhi, Mehedi Masud, Emad Sami Jaha, and Sahil Verma. "Plant Disease Diagnosis and Image Classification Using Deep Learning." *Computers, Materials & Continua* 71, no. 2 (2022).
- [18] Sutaji, Deni, and Oktay Yıldız. "LEMOXINET: Lite ensemble MobileNetV2 and Xception models to predict plant disease." *Ecological Informatics* 70 (2022): 101698.
- [19] Sujatha, Radhakrishnan, Jyotir Moy Chatterjee, N. Z. Jhanjhi, and Sarfraz Nawaz Brohi. "Performance of deep learning vs machine learning in plant leaf disease detection." *Microprocessors and Microsystems* 80 (2021): 103615.
- [20] G. Endait, A. Nikam, P. Bhorkade, D. G. Lokhande, B. S. Agarkar and P. William, "Hybrid Model to Predict Leaf Disease Prediction Using Ensembling Machine Learning Approach," *2023 World Conference on Communication & Computing (WCONF)*, RAIPUR, India, 2023.